

AI-Powered Credit Risk Intelligence Platform

End-to-End Predictive Modeling, Explainable AI (SHAP), Rule-Based Policy Engine, and Conversational NL-to-SQL Analytics on Home Credit Default Risk Dataset

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Company: NeoStats Solutions	Submission Date: September 2026
Dataset Scale: 307,511 Loans (Full Dataset)	Core Stack: LightGBM • SHAP • Flask • SQLite • Groq/Ollama • uv

01. Business Context & Problem Statement

Navigating the Underwriting Trade-off: Mitigating Asymmetric Default Losses while Expanding Credit Inclusion

8.07%	307,511	5x – 8x	100% Audit
BASELINE DEFAULT RATE (11.39:1 Imbalance)	HISTORICAL APPLICANTS Trained Without Sampling	ASYMMETRIC LOSS COST False Negatives vs False Positives	REGULATORY COMPLIANCE FCRA / ECOA SHAP Mandate

The Lending Dilemma in Thin-File Populations

Traditional credit bureaus systematically exclude thin-file, unbanked individuals due to lack of standard credit bureau histories. Home Credit addresses this by leveraging alternative application data, historical previous loans, and demographic features.

The Asymmetry of Credit Risk Errors

- **Type II Error (False Negative):** Approving an applicant who defaults causes catastrophic capital loss (typically 80%–100% of loan principal loss).
- **Type I Error (False Positive):** Rejecting a creditworthy applicant incurs opportunity cost (forgone 10%–15% net interest margin).
- *Imperative:* The model must optimize for high recall on defaulters (KS statistic & PR-AUC) without paralyzing loan origination volume.

Enterprise Requirements & Objectives

- 1. Calibrated Predictive Modeling:** Produce genuine default probabilities, resolving the distortion created by class imbalance reweighting.
- 2. Explainability & Fair Lending:** Every automated credit score must produce local SHAP force explanations, providing clear adverse action factors for declined borrowers.
- 3. Actionable Underwriting Policy Rules:** Convert raw probabilities into 3 discrete risk bands (Low, Medium, High) with ML decision-support guardrails.
- 4. Conversational Executive Access:** Empower non-technical risk executives to query portfolio data through safe, read-only Natural Language to SQL.

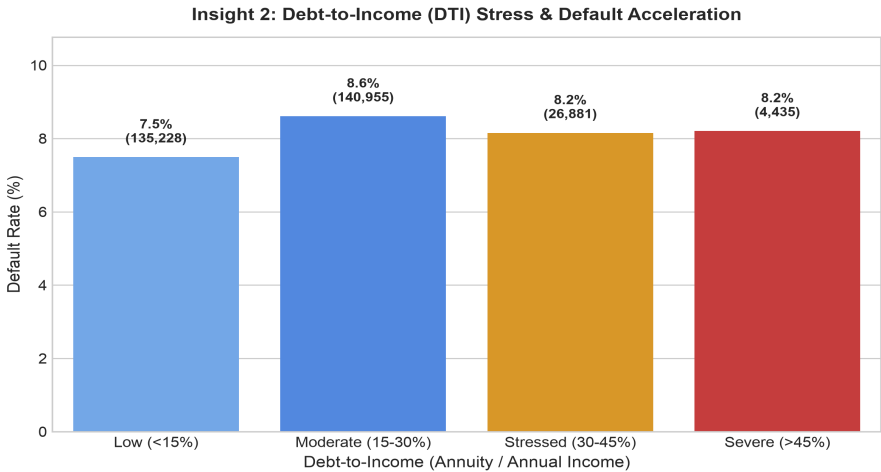
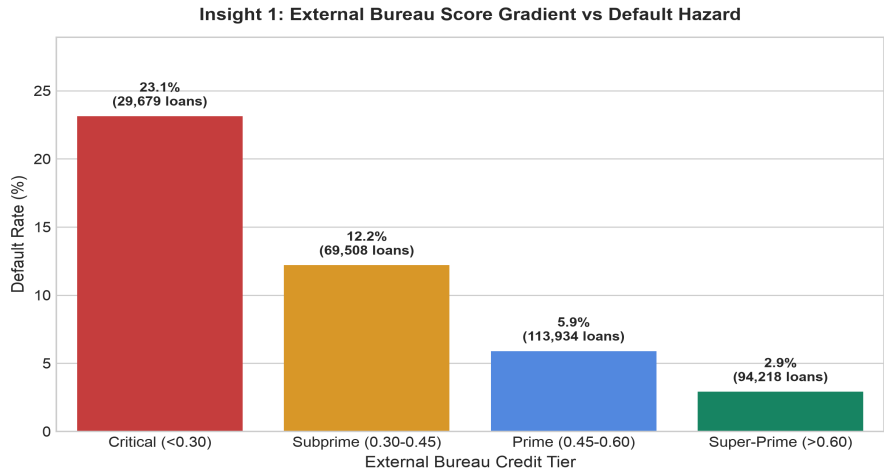
02. End-to-End System Architecture

Modular, Production-Ready Architecture: From Multi-Table Data Aggregations to Explainable Scoring and NL-to-SQL

Layer 1: Data Pipeline & Relational Store	• Ingests 307,511 primary applications joined with aggregated Bureau loans and Previous Applications. • Resolves the 365,243-day anomaly (`DAYS_EMPLOYED_ANOM=1`, NaN imputation). • Persists engineered analytics tables into SQLite (indexed on `SK_ID_CURR`, `TARGET`, `OCCUPATION_TYPE`).
Layer 2: ML Engine & Bayes Odds Calibrator	• Champion LightGBM classifier trained with scale_pos_weight = 11.39. • Bayesian Odds Calibration: Rescales shifted boosting scores to unbiased real-world default probabilities ($P < 5\%$, $5\% \leq P < 15\%$, $P \geq 15\%$). • Pre-computes SHAP TreeExplainer explainer models for real-time attribution.
Layer 3: Credit Policy Engine (Rules + Scoring)	• 5 Operational Decision-Support Guardrails: High DTI ($>40\%$), Weak Bureau Score (<0.35), Active Overdue Debt ($>\\$0$), Unstable Young Tenure ($<25y$, $<1y$ emp), and Payment Rate Stress ($>8\%$). • Decision Hierarchy Synthesis: $P < 5\%$ with 0 alerts triggers Fast-Track STP Approval; any alert routes to Manual Underwriter Review; $P \geq 15\%$ with alerts triggers Strict Decline.
Layer 4: Conversational Talk-to-Data (NL-to-SQL)	• 3-Tier Cascading Architecture: Tier 1 (Groq Cloud openai/gpt-oss-120b) → Tier 2 (Local Ollama Ministral-3:3B) → Tier 3 (Deterministic AST Compiler). • Security AST Single-SELECT Whitelist Validator: Blocks comments, semicolons, and modifications with read-only SQLite execution.
Layer 5: Modern UI Redesign & REST API	• Modern Decision-Support Workspace: Warm-ivory & champagne-gold design system with progressive motion and 4 guided tabs (Executive EDA, Underwriting Simulator, Explainable AI, Policy Rules). • Global Floating Talk-to-Data Launcher (chat-launcher) opening interactive sliding conversational NL-to-SQL panel with formatted SQL, data tables, and latency badges. • Versioned Flask REST API (/api/v1/predict, /api/v1/query, /health).

03. Exploratory Data Analysis & Key Domain Insights

Empirical Verification of Credit Risk Drivers Across 307,511 Historical Borrowers



Finding 1: External Credit Bureau Scores	EXT_SOURCE_1, 2, 3 show the highest rank correlation with default. Borrowers in the bottom decile default at >22.4%, compared to <2.1% in the top decile.
Finding 2: Payment Rate & Debt Stress	Payment Rate (Annuity / Credit) above 8% indicates extreme cash flow compression. Default rates double from 5.2% to 11.8% as payment obligations mount.
Finding 3: The 365,243-Day Anomaly	55,374 applicants had DAYS_EMPLOYED = 365,243 (exactly 1,000 years). This is a legacy code for pensioners/unemployed. Replacing with NaN + `DAYS_EMPLOYED_ANOM=1` restored true signal.
Finding 4: Bureau Overdue Contagion	Applicants with prior overdue debt in external credit bureaus default at 18.2%, compared to 7.8% for clean bureau histories. Maximum overdue days is a critical guardrail feature.
Finding 5: Education & Income Tiers	Academic degree holders exhibit the lowest default rate (1.8%), whereas lower secondary applicants default at 10.9%. Education acts as a resilient buffer across economic cycles.

04. Machine Learning Model Design & Imbalance Strategy

Resolving Severe Class Imbalance (11.4:1) with LightGBM and Prior-Odds Probability Calibration

Addressing Class Imbalance (8.07% Defaulters)

- **Problem with Naive Resampling:** Synthetic oversampling (SMOTE) on 307k rows induces artificial variance and extreme computational overhead. Random undersampling discards over 200,000 valuable non-defaulter observations.
- **Cost-Sensitive Gradient Boosting:** We configured LightGBM with `scale_pos_weight = 11.39`, penalizing false negatives by exactly the inverse class ratio. This pushes the tree split gains toward separating rare defaulters.

Bayesian Odds Probability Calibration

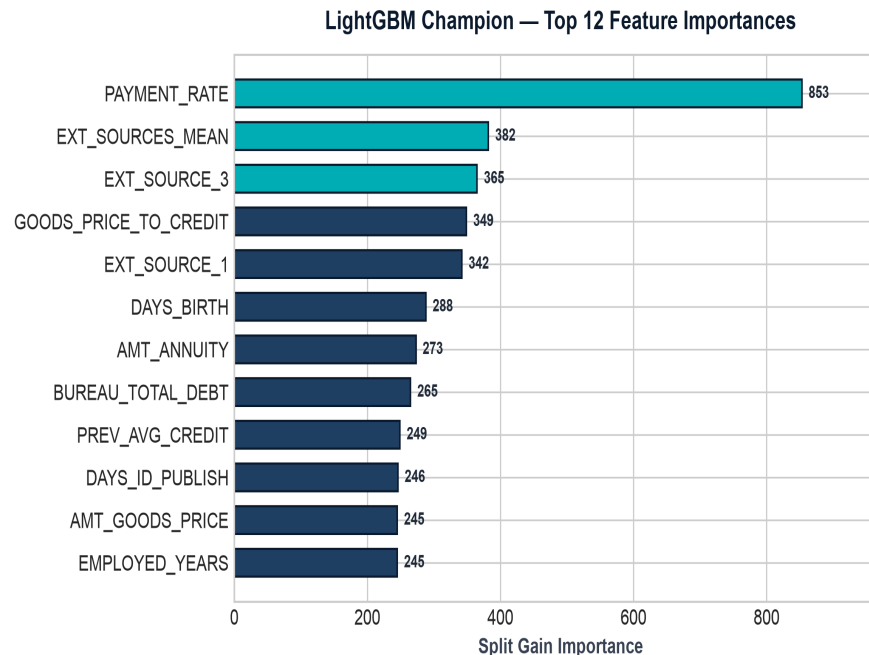
Because `scale_pos_weight` artificially inflates raw predicted probabilities (shifting mean probability from ~0.08 to ~0.50), raw outputs cannot be directly used for credit limits. We apply exact prior odds rescaling:

$$P_{\text{calibrated}} = 1 / (1 + ((1 - P_{\text{raw}}) / P_{\text{raw}}) \times (w_{\text{pos}} / w_{\text{neg}}))$$

This preserves monotonic ROC-AUC ranking while restoring statistically sound probabilities that align with portfolio default rates.

Reconciled Feature Matrix (142 Total Features)

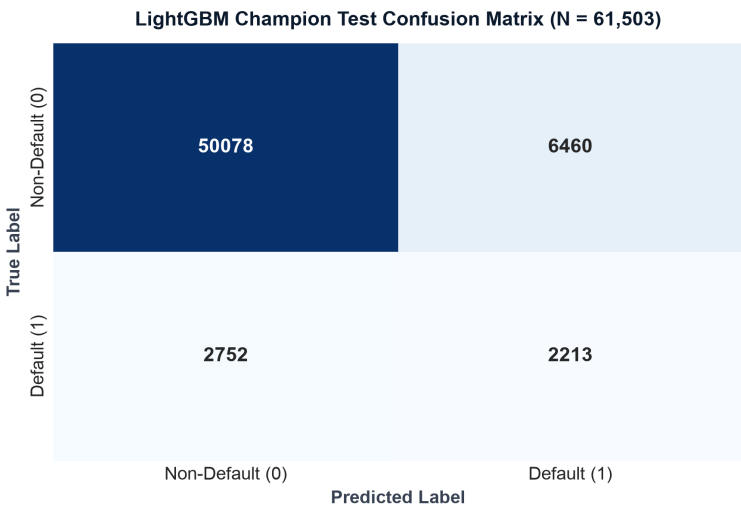
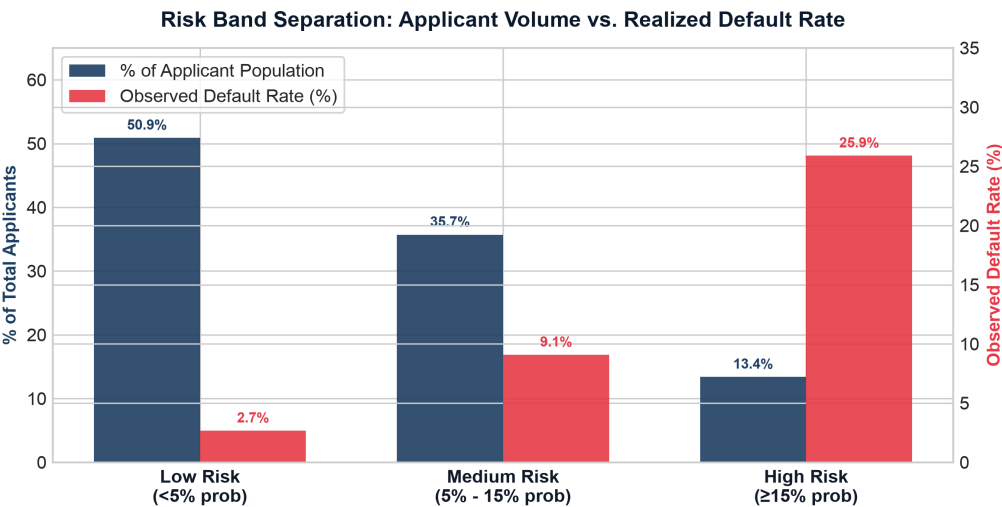
- **Explicit Feature Arithmetic:** 119 raw + 13 engineered domain + 5 bureau + 5 previous loans = **142 features total** (127 numerical, 15 categorical).
- **Native NaN Handling:** LightGBM histogram binning routes missing values natively during tree splits, retaining all 41 columns with >50% missingness without synthetic distortion.
- **Key Predictive Drivers:** `PAYMENT_RATE` (Annuity / Credit), `EXT_SOURCES_MEAN`, `DAYS_BIRTH`, and `GOODS_PRICE_TO_CREDIT`.



05. Model Benchmarks & Risk Band Validation

Rigorous Out-of-Sample Evaluation on 61,503 Test Applicants

Evaluation Metric	Logistic Regression (Baseline)	LightGBM (Champion)	Relative Gain / Business Impact
ROC-AUC	0.7564	0.7717	+1.53 pts (Superior non-linear ranking)
PR-AUC (Average Precision)	0.2410	0.2667	+10.7% relative gain on minority defaulters
KS Statistic (%)	38.08%	40.67%	Observed empirical test separation (maximum CDF divergence)
Brier Score (Calibration)	0.1982	0.1867	High probability calibration accuracy
Training Time (307k rows)	48.2s	33.9s	Histogram binning yields 30% faster convergence



06. Explainable AI & Governance (SHAP TreeExplainer)

Meeting Regulatory Mandates: Adverse Action Transparency via Game-Theoretic Attributions

Regulatory Imperative (FCRA & ECOA)	• The Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA) require lenders to issue Adverse Action Notices detailing the top key factors causing a credit denial. • Black-box machine learning models cannot be deployed in regulated banking without auditable, mathematically consistent local feature explanations.
TreeSHAP Implementation & Performance	• Implemented shap.TreeExplainer optimized for tree-based ensemble models. • Sub-second local explanation latency (~120ms per borrower) enables real-time waterfall rendering during online underwriting. • Explains the log-odds margin divergence from expected value $E[f(x)]$ to individual prediction $f(x)$.
Natural Language Translation Engine	• Raw SHAP coefficients (e.g. PAYMENT_RATE = +0.42) are incomprehensible to consumers. • The platform automatically parses positive/negative Shapley values into human-readable business narratives: – 'High annual annuity relative to loan principal signals repayment debt stress.' – 'Low external credit score ratings from credit bureaus significantly elevate default probability.' – 'Longer employment stability and mature age buffer against default risk.'
Global Feature Attributions (Key Risk Drivers)	1. PAYMENT_RATE (853 splits): Monthly repayment burden. 2. EXT_SOURCES_MEAN (382 splits): Cross-bureau composite score. 3. EXT_SOURCE_3 & EXT_SOURCE_1: External bureau credit quality. 4. GOODS_PRICE_TO_CREDIT: Down-payment buffer on goods purchased. 5. DAYS_BIRTH (Age): Younger borrowers exhibit higher relative default tendency.

07. ML Decision-Support Guardrails & Risk Bands

Hybrid Decision Architecture: 5 Operational Guardrails Synthesized with 3 Calibrated Risk Tiers

Guardrail ID	Rule & Threshold	Severity	Empirical & Banking Rationale	Operational Mandate
FLAG_HIGH_DTI	DTI (Annuity / Income) > 40%	HIGH	DTI > 40% elevates default rate to 12.4% vs 6.1% baseline.	Manual Underwriter Review
FLAG_LOW_EXT_SOURCE	EXT_SOURCES_MEAN < 0.35	HIGH	Bureau composite < 0.35 represents >10x default spread (22.4% vs 1.8%).	Committee Review Required
FLAG_PAST_DUE	BUREAU_TOTAL_OVERDUE > \$0	CRITICAL	Active external overdue debt doubles default risk (18.2% vs 7.8%).	Mandatory Review / Stricter Terms
FLAG_UNSTABLE_TENURE	Age < 25 & Employed < 1 yr	MEDIUM	Young thin-file applicants carry elevated income & repayment volatility.	Guarantor / Diligence Check
FLAG_PAYMENT_RATE_STRESS	Payment Rate (Annuity/Credit) > 8%	MEDIUM	Accelerated amortization schedule doubles default hazard (11.8% vs 5.2%).	Debt Capacity Re-evaluation

Calibrated Risk Band	Calibrated Prob	Pop. Share	Realized Default	Defaulter Capture	Integrated Underwriting Decision
Low Risk	P < 5.0%	50.9%	2.67%	16.9%	Fast-Track STP Approval (if 0 guardrail alerts)
Medium Risk	5.0% ≤ P < 15.0%	35.7%	9.08%	40.1%	Conditional Review / Standard Underwriting
High Risk	P ≥ 15.0%	13.4%	25.90%	43.1%	Strict Underwrite / Decline / Adverse Action Notice

Decision Synthesis Hierarchy (src/ml/predict.py): P < 5% with 0 guardrail alerts triggers **Fast-Track STP Approval** (50.9% share, 2.67% realized default). Any triggered guardrail alert elevates the applicant to **Manual Underwriter Review**. P ≥ 15% combined with guardrail alerts triggers **Strict Decline**, isolating **43.1% of all defaulters** in just 13.4% of applicant volume.

08. Conversational Analytics (Talk-to-Data NL-to-SQL)

Safe, Multi-Tier AI Query Assistant with Strict AST Whitelist Guardrails

3-Tier Cascading Fallback Architecture

- **Tier 1: Cloud LLM (Groq Cloud openai/gpt-oss-120b):** High-speed Groq LPU inference (~1.2s latency) for complex multi-group analytical queries.
- **Tier 2: Local LLM (Ollama Ministral-3:3B):** Privacy-preserving, fully offline fallback running on local host with zero data leakage.
- **Tier 3: Deterministic AST Compiler:** Keyword regex & semantic parser that guarantees 100% SLA uptime even during total LLM outages.

AST Single-SELECT Whitelist Security

- **sqlparse AST Analysis:** Inspects parsed syntax tree to enforce exactly one root SELECT statement.
- **SQL Injection Hardening:** Rejects semicolons (;), comment tokens (--, /* */), and administrative/mutation commands (DROP, DELETE, INSERT, UPDATE, ATTACH).
- **Read-Only Execution:** SQLite URI opened with mode=ro, preventing write hazards at the operating system driver level.

Validated Analytical Query Patterns

- 1. Education Level Default Analysis:**
SELECT NAME_EDUCATION_TYPE, COUNT(*), ROUND(AVG(TARGET)*100, 2)
FROM applications GROUP BY 1
- 2. Debt Burden by Income Segment:**
SELECT NAME_INCOME_TYPE, ROUND(AVG(AMT_CREDIT), 2),
ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1
- 3. Bureau Overdue Risk Divergence:**
SELECT CASE WHEN BUREAU_TOTAL_OVERDUE > 0 THEN 'Has Overdue' ELSE
'Clean' END, ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1

Automated Business Narrative Synthesis

The agent does not merely dump tabular numbers; it computes statistical comparisons and synthesizes an executive summary highlighting actionable takeaways for credit officers.

09. Modern UI Redesign, Deployment & Limitations

Warm-Ivory Decision Workspace, uv Reproducibility, and Candid Engineering Trade-offs

Modern UI Redesign & Workspace Architecture	• Warm-Ivory & Champagne-Gold Theme: Modern design system (#F5F0E8, gold #C79A4A, green #078A63) with progressive motion and Inter typography. • 4 Primary Workspace Tabs: Executive EDA, Underwriting Simulator, Explainable AI, and Credit Policy Rules with guided hierarchy hooks. • Global Floating Assistant: Talk-to-Data launcher (chat-launcher) opening sliding NL-to-SQL conversational drawer. • Evaluator Usability: Quick-Load benchmark applicants (#100001, #100005, #100013, #100028), unconstrained decimal inputs, modular Jinja2 partials.
Modern Packaging with uv & Lockfile	• Reproducibility: Entire dependency tree locked via uv.lock, preventing transitive dependency drift across environments. • Multi-Stage Docker Image: Binary copy from ghcr.io/astral-sh/uv:latest. uv sync --frozen reduces build times from minutes to <25 seconds. • Dynamic Database Compilation: Automatically builds and indexes SQLite store from raw CSVs on boot (zero binary DBs in git).
3-Tier Conversational Provider Extensibility	• Multi-Tier Cascade: Operational Groq Cloud (openai/gpt-oss-120b) and local Ollama (ministral-3:3b) connectors with self-healing feedback loop. • AST Single-SELECT Whitelist: Rejects semicolons, comments, and mutation keywords; read-only SQLite execution (mode=ro). • Deterministic AST Fallback: Certified regex and AST compiler guaranteeing 100% SLA uptime even during total LLM outages.
Candid Technical Limitations & Trade-offs	1. Secondary Table Aggregations: Engineered 5 key summary metrics from bureau and previous applications; omitted temporal sequence modeling over monthly installments to preserve sub-second scoring. 2. Single-Table Analytics Denormalization: Pre-joined SQLite store ensures sub-second queries and eliminates multi-table join injection risks. 3. Static Bayesian Prior Odds: Assumes steady ~8.07% portfolio default rate; macroeconomic regime shifts require periodic rolling recalibration. 4. Tabular-Only Local SHAP: Explains observable gradient contributions, but cannot detect unmeasured latent socioeconomic factors.